

ASCII MASTER FLOW – PANEL 1/12: Transport to trade chain

MINE OR FARM -> production point with dispersed origins

ROADHEAD -> flexible collection over short hauls

RAIL OR WATERWAY CORRIDOR -> bulk trunk haul over long distance

PORT OR AIRPORT -> break of bulk and interface with global routes

WORLD MARKET -> exchange completed through connectivity, cost and time

MUST REMEMBER: Transport geography explains networks through nodes, links, accessibility, connectivity, density, hierarchy, break-of-bulk, modal cost and hinterland; trade adds comparative advantage, corridors, ports, logistics and chokepoints, while India's space programme links launch, satellite, application and governance institutions.

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ASCII MASTER FLOW – PANEL 2/12: Modal cost structure comparison

ROAD -> low terminal cost, high per-km cost; short hauls and door-to-door delivery

RAIL -> higher terminal cost, lower per-km cost; medium to long bulk hauls

WATER -> highest terminal cost, lowest per-km cost; long bulky low-value cargo

AIR -> costly throughout; justified by high value-to-weight, urgency or inaccessibility

PIPELINE -> very high fixed cost, very low operating cost; continuous liquid or gas flow

ASCII MASTER FLOW – PANEL 3/12: Modal choice decision rule

IDENTIFY CARGO -> bulk, value density, urgency and perishability

IF haul is short -> terminal cost dominates, road usually wins

IF haul exceeds the road-rail break-even -> rail or water recovers its terminal cost

IF value-to-weight is very high or access is difficult -> air becomes defensible

ALWAYS -> treat freight as an intermodal chain, not a single modal verdict

ASCII MASTER FLOW – PANEL 4/12: Network vocabulary rail

HINTERLAND -> inland area served by a port, city or route

NODE -> junction or terminal where flows converge

CORRIDOR -> linear axis of intense transport and economic activity

BREAK OF BULK -> port, railhead or dry port where cargo changes mode

HUB AND SPOKE -> one major hub redistributes flows to smaller nodes

ASCII MASTER FLOW – PANEL 5/12: Port site versus situation

SITE -> depth, shelter, tidal range, foreshore, expansion land, siltation risk

SITUATION -> hinterland productivity, inland connectivity, position on shipping lanes

CORRECTABILITY -> site can be dredged and engineered; situation cannot

GROWTH RULE -> situation normally decides which ports grow

POLICY -> freight corridors and inland container depots are port instruments

ASCII MASTER FLOW – PANEL 6/12: Port type worked comparison

NATURAL HARBOUR -> excellent site; stays small if the hinterland is poor

ESTUARINE PORT -> good route situation, but siltation and draught limit the site

ARTIFICIAL DEEP-WATER PORT -> engineered site chosen for hinterland reach and rail links

SATELLITE CONTAINER TERMINAL -> deep draught and back-up land beside a congested parent

TRANSHIPMENT HUB -> situation is the route itself, so it lives without a local hinterland

ASCII MASTER FLOW – PANEL 7/12: Containerisation hierarchy effect

STANDARD CARGO UNIT -> handling time and cost collapse at every transfer

INTERMODAL TRANSFER BECOMES ROUTINE -> production fragments across countries

ADVANTAGE SHIFTS -> deep draught, crane productivity and inland rail connectivity

TRAFFIC CONCENTRATES -> fewer, larger hubs with feeder services to the rest

ANSWER USE -> a hierarchy argument that needs no throughput statistic

ASCII MASTER FLOW – PANEL 8/12: Chokepoint vulnerability rail

NARROW PASSAGE -> world shipping is funnelled through a small cross-section

DISRUPTION -> voyages lengthen and rerouting begins immediately

COST TRANSMISSION -> freight and insurance rise far beyond the affected region

POLICY RESPONSE -> chokepoint security, alternative routes and canal capacity

READING -> the passage is valuable as route control, not as a water body

ASCII MASTER FLOW – PANEL 9/12: Transport and development two-way fork

NEW LINK OPENS -> relative accessibility of every node changes, not only the two joined
IF junctions, local loading and complementary investment exist -> corridor growth follows
IF they are absent -> the tunnel effect carries benefit to the endpoints
IF local producers are weak -> backwash exposes them and speeds out-migration
VERDICT -> connectivity is necessary but not sufficient for regional development

ASCII MASTER FLOW – PANEL 10/12: India multimodal corridor ladder

BHARATMALA -> economic corridors, inter-corridors, border and port connectivity

DEDICATED FREIGHT CORRIDORS -> separate freight from passenger congestion on major axes

SAGARMALA -> port modernisation, port-led industrialisation and coastal connectivity

PM GATI SHAKTI -> network integration replacing mode-wise sectoral planning

TRADE MAP -> western container and petroleum gateway; eastern bulk and mineral coast

CLOSE DISTINCTION: Road/rail density differs from effective accessibility, port capacity from throughput, inland waterway declaration from operational traffic, trade balance from current account, ISRO from the Department of Space and NewSpace India Limited, and navigation/communication/earth-observation missions retain distinct functions.

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ASCII MASTER FLOW – PANEL 11/12: Space as applied geography

EARTH OBSERVATION -> land use, crop condition, forests, water bodies, coasts, glaciers

METEOROLOGY -> cyclone tracking and monsoon monitoring feeding warning systems

NAVIGATION AND COMMUNICATION -> positioning, timing and remote-region connectivity

GIS INTEGRATION -> imagery plus terrain, infrastructure and socio-economic layers

OUTPUT -> site selection, corridor alignment, watershed planning, damage assessment

EVIDENCE LIMIT: Golden Quadrilateral, Dedicated Freight Corridors, Bharatmala, Sagarmala, major ports, National Waterway-1, Delhi-Mumbai corridor, Malacca/Hormuz/Suez and PSLV/GSLV/LVM3, INSAT/GSAT, IRS/EOS and NavIC form named map-institution anchors; project and mission claims require official date/status qualification.

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ASCII MASTER FLOW – PANEL 12/12: Space institutions and the operational-orbit caution

ISRO -> national space architecture organised around application, not prestige
SRIHARIKOTA -> east-coast launch site with over-sea trajectory safety advantage
NSIL AND IN-SPACe -> commercialisation and non-governmental participation
NavIC -> indigenous navigation supporting transport timing and positioning
NVS-02 ON 29 JAN 2025 -> reached transfer orbit but not its intended NavIC slot